Definition of a species:

What is monoculture?

What does biodegradable mean?

What is sustainability?

Draw a diagram of the carbon cycle

Include these words:

- photosynthesis
- respiration (twice)
- decay
- combustion
- decomposing

Extinction rates are increasing due to:

New species evolve due to:

Evolution and ecology

What is a habitat?

Using similarities and differences between living things to put them into groups is called:

Describe the process of natural selection:

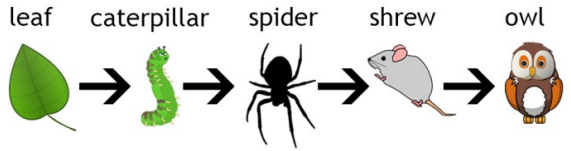
Look at the diagram below and name a:

producer: predator:

What might the shrews be competing for?

Who would be at risk if the caterpillars died out?

This diagram is called a:



We can change the features of a species by:

The two scientists with different theories about the changes in species were:

New species form when their environment changes. This happens because of:

Nitrogen cycle involves the processes of:

What is the definition of extinction?

What's the difference between local and mass extinction?

Why do we need biodiversity?

What is an adaptation?

Give examples of an adaptation to:

a. hot place

b. cold place

Scientists measure environmental changes by looking at indicator organisms.

Increase in fertiliser run off leads to an increase in -

These animals are only present in clean water -

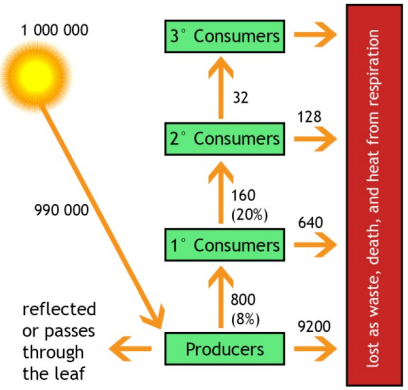
Plant sensitive to air pollution -

Carbon cycle involves the processes of:

Evolution and ecology

Name for the process by which species change over time:

Not all of the energy at each stage of the food chain is passed on. What happens to the rest of it?



What is the ultimate source of energy?

Evidence for evolution comes from:

F .....

D .....

Suggested answers

Definition of a species:

A group of organisms that can produce fertile offspring

What is monoculture?

Cultivation of a single crop

Extinction rates are increasing due to:

Human activity causing increasing rates of habitat loss and climate change

Look at the diagram below and name a:

Producer: *plant*
Predator: *spider/shrew/owl*

What might the shrews be competing for? *Food - spiders, space, water*

Who would be at risk if the caterpillars died out? *The rest of the food chain – spiders/shrews/owls*

This diagram is called a: *food chain*

Not all of the energy at each stage of the food chain is passed on. What happens to the rest of it?

Lost in waste materials / heat energy

Energy released by respiration

What does biodegradable mean?

Can be decomposed by microorganisms

What is sustainability?

Maintaining a balance by avoiding depletion of natural resources

Describe the process of natural selection:

Variation within a species, individuals with favourable characteristics survive to reproduce and pass genes onto next generation

Why do we need biodiversity?

For global food security, increased gene diversity reduces risk of disease, development of medicines etc

Using similarities and differences between living things to put them into groups is called:

Classification

What is a habitat?

The natural environment in which an organism usually lives

Evolution and ecology

Not all of the energy at each stage of the food chain is passed on. What happens to the rest of it?

Lost in waste materials / heat energy

Energy released by respiration

The two scientists with different theories about the changes in species were:

Lamarck, Darwin/Wallace

New species form when their environment changes. This happens because of:

Natural selection

Name for the process by which species change over time:

Evolution

We can change the features of a species by:

Natural selection

What is the ultimate source of energy?

The Sun

Scientists measure environmental changes by looking at indicator organisms.

Increase in fertiliser run off leads to an increase in *algae.*

These animals are only present in clean water – *mayfly larva/freshwater shrimp*

Plants sensitive to air

What is the definition of extinction?

The dying out of a species

What's the difference between local and mass extinction?

Mass extinction is a period when unusually large numbers of species die out whereas local extinction is when a species ceases to live in an area but still

Nitrogen cycle involves the processes of:

Nitrogen

Fixation

Decay

Denitrification

What is adaptation?

A characteristic that has developed over time that allows an orgaism to live successfully in its habitat.

Give examples of an adaptation to:

a. hot place: cactus – *leaves reduced to spines to reduce water loss*

New species evolve due to:

Mutation, genetic variation, isolation, natural selection, speciation

Carbon cycle involves the processes of:

Photosynthesis

Respiration

Combustion

Decay